



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Driven Digital Twin	
S. No.	University ID	Name
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Abstract

Students often struggle to understand how their study habits, attendance, academic performance, and daily activities influence their overall results. The AI-Driven Digital Twin for Personalized Student Performance Prediction and Adaptive Learning is an innovative software platform designed to create a virtual representation of each student and continuously analyze their academic and behavioral data using Artificial Intelligence. The proposed system collects and analyzes student information such as study hours, attendance, quiz marks, assignment completion, and optionally sleep patterns. The digital twin continuously updates based on the student's latest activities and performance. AI and Machine Learning techniques are used to predict academic performance, identify the risk of failure, analyze performance trends, and provide personalized recommendations for improving learning outcomes. The system also supports what-if analysis, allowing students to understand how changes in their study habits, attendance, or assignment completion may affect their predicted performance. By combining Digital Twin technology with Artificial Intelligence and predictive analytics, the system provides students with an interactive representation of their academic progress. The proposed solution follows the principles of Adaptive Software Engineering by continuously adapting the digital twin according to changes in student behavior and performance. It aims to improve academic decision-making, identify potential academic risks at an early stage, encourage effective study habits, and provide personalized learning recommendations. The project demonstrates how modern AI technologies and Digital Twin concepts can be integrated to create an intelligent, adaptive, and personalized academic support system for students

Signature of the Guide

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